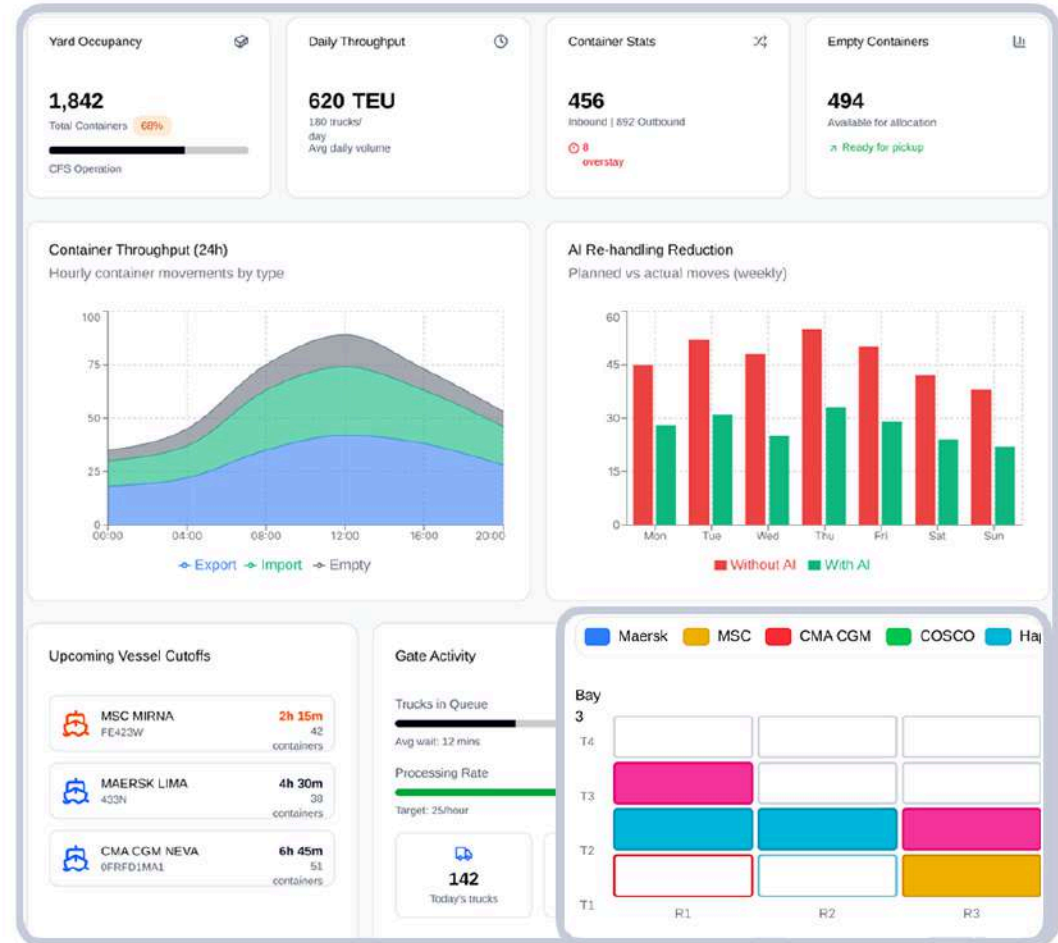




Yard Optimize

Every re-handle begins as a placement decision.

Powered by **Ctrl + Stack**



Port Terminals & Container Handling



Sea Side Yards

Land Side Yards

Problem Statement



Re-handling begins before the first move.

Containers are stacked without clear visibility into exit intent, priority, or future congestion.

Influenced by vessel cut-offs, truck arrivals, yard saturation, priority shifts, equipment availability, and stacking rules.

The Result:

- ⚠ Excessive Rehandling
- ⚠ Congested Yards
- ⚠ Idle Equipment and Labour
- ⚠ Delayed Throughputs



Solution Overview



1. Yard Twin

- Real-time Block-Bay-Row-Tier visualization

2. Smart Slotting Engine

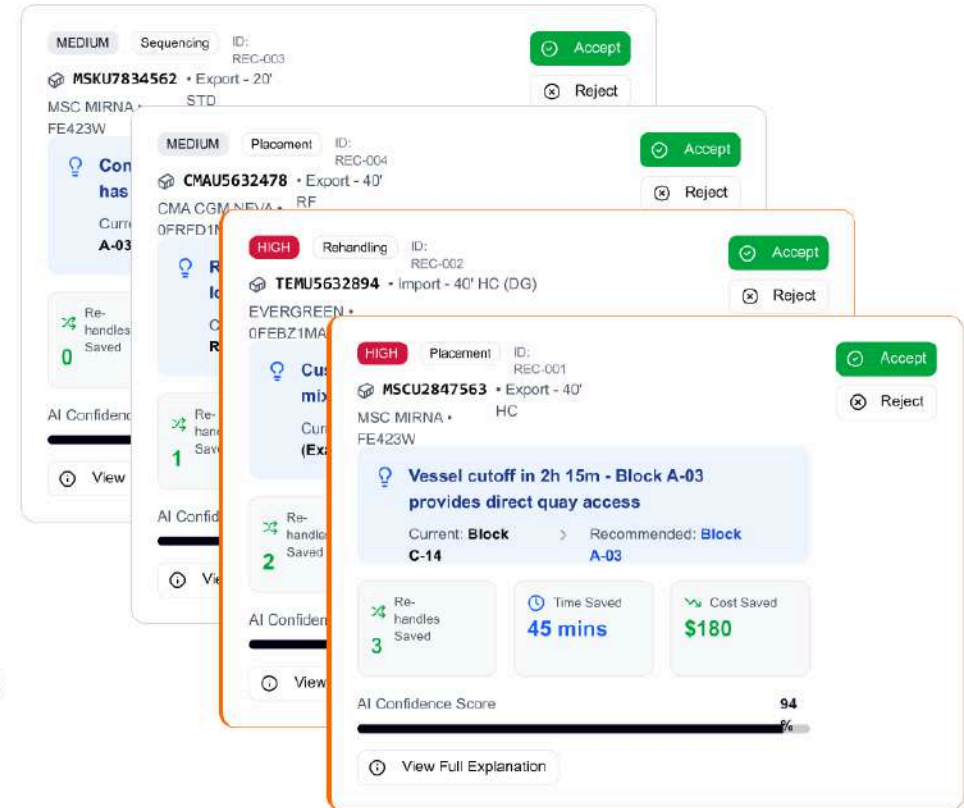
- AI-recommended stack location on yard-in
- Considers cutoff, priority, weight, hazards
- Minimizes future reshuffles & crane travel

3. Rule & Constraint Engine

- Hard & soft rules - with varying priorities
- Configurable & explainable decisions

4. Retrieval & Resequencing Orchestrator

- Auto-plans restacks with AI-suggested locations
- Guides gate-out & vessel staging decisions



Hackathon Prototype



Dashboard

Real-time operational overview with key metrics, yard utilization, and performance indicators across all terminal operations



Configuration

System settings for yard structure, stacking rules, business constraints, handling equipment, and terminal-specific operational parameters



Yard View

Interactive 3D visualization of container placement across all yards (Sea-Side, Land-Side, Special Zones) with bay-row-tier slot management



Container Movement

Complete container lifecycle management from gate-in to retrieval, handling incoming cargo, yard inventory, and export sequencing



Recommendations AI-Powered

Sequenced AI-powered optimization suggestions with explainable insights and star markers indicating constraint-based placement decisions



AI Assistant

Context-aware conversational assistant with Pointer/Inspect mode, document autofill from Shipping Bills/ Bills of Lading, persistent chat across navigation, and actionable recommendations

📍 SS3-B4-01-03-2

Placement Metrics & Analysis

Placement Score 98%

Excellent - Highly optimized location

🔄 Rehandle Risk LOW

15% chance of requiring rehandle

🕒 Retrieval Efficiency MEDIUM

Estimated retrieval: 7 min

📍 POD Clustering 94

Excellent - Strong POD clustering

⚖️ Weight Compliance ✓ COMPLIANT

40.1 MT (Heavy) → Tier 2

📏 Block Utilization 37

Low - Plenty of space

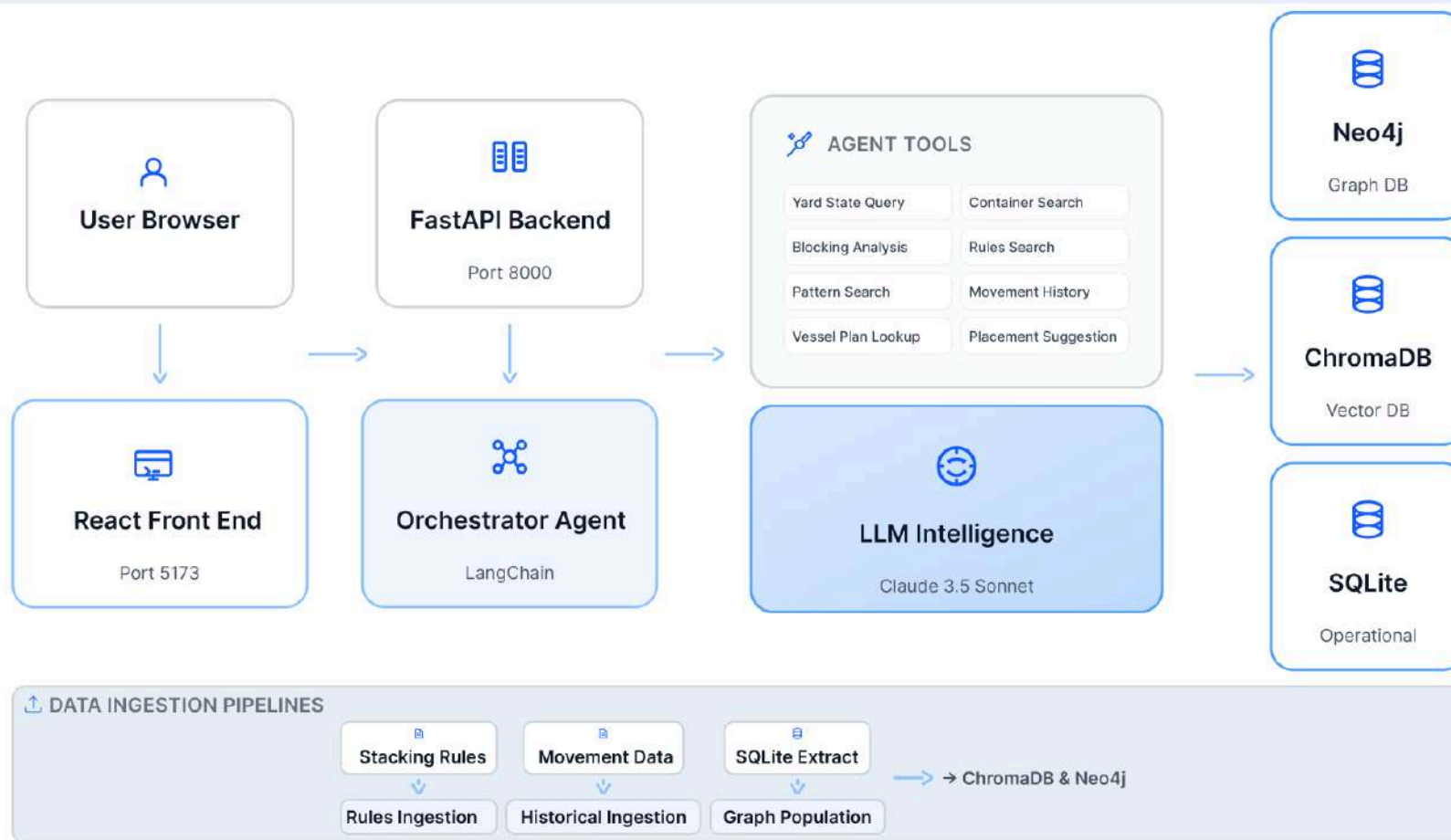
🚢 Distance to Quay 101m

🕒 Est. Retrieval Time 7 min

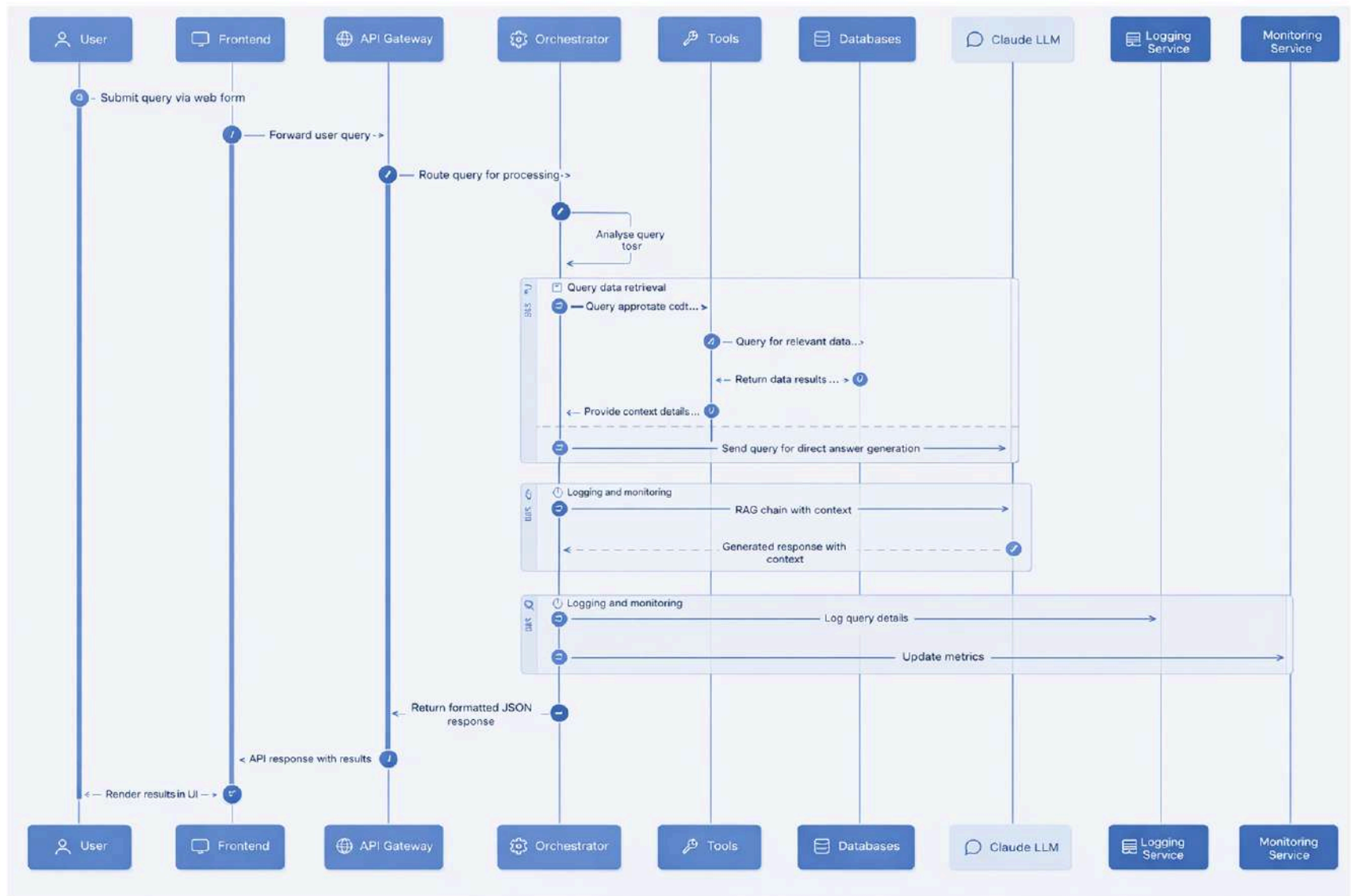
📦 Blocking Containers 2

📏 Congestion Level Low

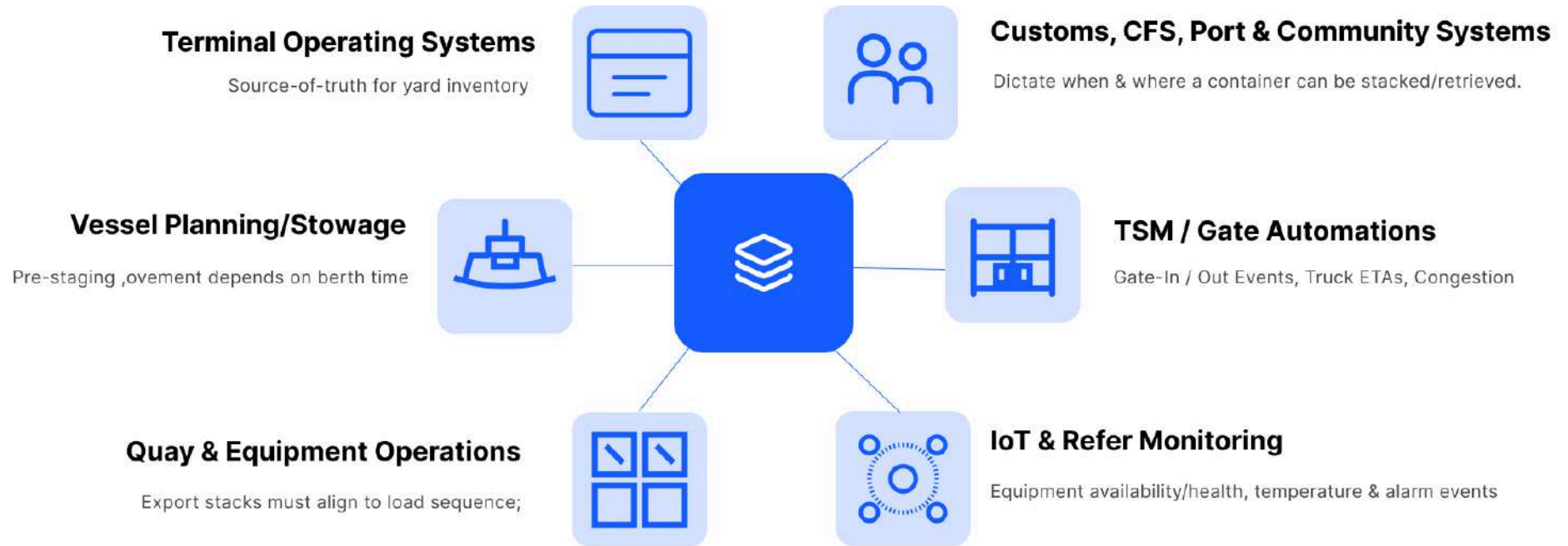
System Architecture



Data Query Processing Flowchart



System Integrations



Product Vision



To transform container yards from ~~reactive storage spaces~~ into **intelligent, self-optimizing systems**.



1 First-Move Right Placement

Get placement right the first time—avoid reshuffles later



2 Intelligence & Automation

AI guidance that accelerates planner decisions



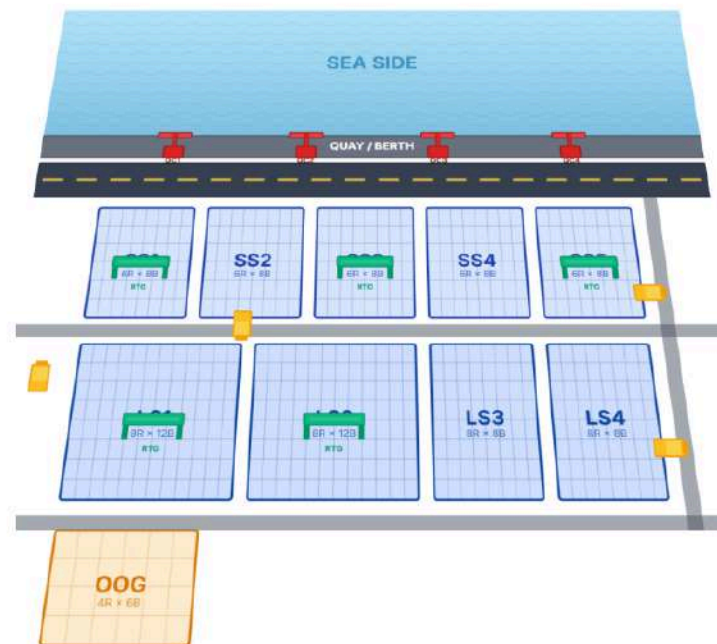
3 Explainability & Auditability

Every move explained. Every decision traceable.



4 Scalability & Modularity

Built to scale across yards, terminals, and models



Minimum Desirable Product



Port Ready: Designed for safe, phased deployment with human control and operational guardrails



Data Readiness

Live container master, yard topology, real-time container movement events



Operational Rules

Hard rules, override hierarchy, and exception handling for safe moves



Security & Ops

Role-based access, audit logs, and fail-safe / AI-off modes



Intelligence Layer

Stack-time placement, retrieval feasibility checks, and safe fallback locations



User Experience

Live yard twin, container detail view, and mandatory override reasoning



Integrations

Read-only integration with TOS, CFS/Depot systems, TAMs, and vessel cut-off feeds



Analytics

Rehandles, forced vs planned moves, and basic yard efficiency signals



Event Consistency & Data Quality Checks

Event ordering, duplication handling, and stale data detection

Product Roadmap



MDP (0-3 Months): Safe, controlled deployment ready for live terminal operations

Phase 1 (6 Months): Prove value through pilots and refined intelligence

- **Prototype Refinement** – Faster AI, higher placement accuracy
- **Pilot Deployment** – Shadow mode rollout with partner terminal
- **Risk Signals** – Rehandle risk, congestion and cutoff warnings
- **Notifications v1** – Planner alerts and operational warnings
- **Analytics v1** – Rehandles, forced vs planned, dwell basics
- **Billing Readiness** – Usage metering and tenant-level tracking

Phase 2 (12+ Months): Scale into a full optimization platform

- **Predictive Optimization** – Proactive re-stacking and congestion avoidance
- **Multimodal Coverage** – Rail, road, vessel-aligned planning
- **EDI Support** – Standard EDI flows (CODECO, COARRI, BAPLIE)
- **Notifications v2** – Role-based, configurable, predictive alerts
- **Billing & Pricing** – Subscription plans, volume-based pricing
- **Enterprise SaaS** – Multi-terminal scale, compliance, SLAs

The Ask:

- *Data Access*
- *Pilot Partners*
- *Mentorship*
- *Resourcing*

Impact Metrics



Cost Reduction

20 - 30 %

Reduction in yard operating cost (fuel/energy + equipment hours + labor time)



Time Saved

25 - 40 %

Improvement in truck turn-around time (TAT) + reduction in container retrieval delays



Accuracy Improved

30 - 50 %

Reduction in misplacements + fewer planning deviations (wrong block/stack, unexpected events)

Scalability & Business Value



Global Scale

Works for any port, anywhere

- Built & validated on 1 yard
- Modular AI adapts to any port size
- Geography & segment agnostic
- Scales to global terminal groups



Business Model

Smart SaaS economics

Revenue Streams

- SaaS subscription per terminal/yard block
- Usage-based: \$0.15~0.50 per container move
- Enterprise licensing for global operators
- Premium: AI planning + KPI dashboards

Unit Economics & EBITDA Path

- Target gross margin: 75–85%
- Low CAC: Direct to terminal operators
- High retention: Mission-critical ops tool
- Rule of 40 target: Growth + margin $\geq 40\%$
- Path to 20%+ EBITDA margin at scale

Recurring Revenue + High Margins



Value Proposition

Why customers pay

- 20~40% reduction in unproductive moves
- Lower fuel & equipment wear costs
- Faster truck & vessel turnaround
- Reduced congestion & demurrage
- ROI typically within 6–12 months



System Integration

We make existing systems smarter

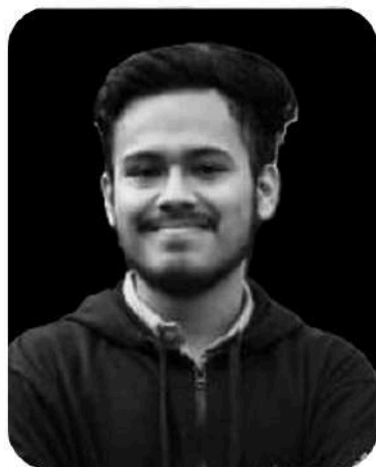
- API-based, event-driven architecture
- Consumes real-time container events
- Outputs optimized stacking & retrieval
- Integrates with Navis, Tideworks, CyberLogitec
- Works with RTGs, reach stackers, cranes
- No system replacement required



Target Customers

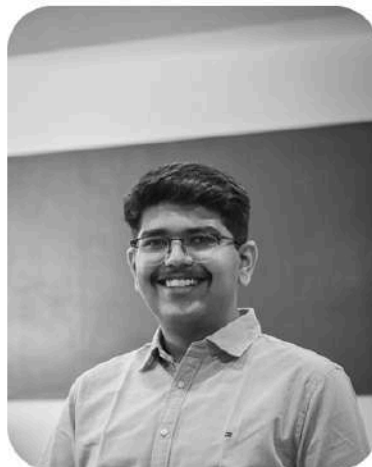
- Container terminal operators
- Port authorities
- Logistics park operators
- Global groups: PSA, DP World, APM

Meet the **Ctrl + Stack** Team



Ayush Sharma

Student
IIT Mumbai



Atharv Ganla

Jr. Data Scientist
ChiStats



Keval Shikhaliya

Software Developer
Kale Logistics



Sahana Ganesh

Sr. Product Manager
Kale Logistics



Shailaja Devade

Team Lead
Kale Logistics



Thank You!

Container Movement

All 10 Export 4 Import 5 Empties 1 Columns

Container No	Current Location	Flow	Source	Container	Actions
MSCU1234567	On Vessel - MSC MAY	Import			
MAEU9876543	In Transit	Export			
CMAU3456789	On Vessel - CMA CGM TROCADEF	Import			
TCLU4567890	At Terminal - Gate 3	Export			
HLCU2345678	On Vessel - HAPAG LLOYD EXPRI	Import			
OOLU5678901	In Transit	Export			
COSU7890123	On Vessel - COSCO PACIF	Import			
EISU8901234	At Terminal - Gate 5	Export			
YMLU9012345	In Transit	Empty			
ZIMU0123456	On Vessel - ZIM MEDITERRANE	Import			

AI Assistant
Container Movement Help

Welcome to AI Assistant

I can help you with managing container movements and optimizing workflows.

Explain AI recommendation Check compliance

Optimize workflow

Ask me anything...

Powered by **Ctrl + Stack**



Appendix

Competitor Analysis



Category	Product / Company	Core Offering	Strengths	Gaps vs Yard Optimize
Traditional TOS	Navis (N4) Full-scale Terminal Operating System	Full-scale Terminal Operating System	Deep operational coverage, widely adopted • Rule-heavy, reactive planning, limited AI explainability, hard to retrofit	
Tideworks	Tideworks TOS + yard planning modules	TOS + yard planning modules	Strong execution workflows • Optimization is static, limited predictive intelligence	
TBA Group	TBA Group Autostore, planning & execution	Autostore, planning & execution	Automation-first terminals • Not modular, high deployment cost, limited human-in-loop	
Planning & Optimization Tools	Portchain Berth & port call optimization	Berth & port call optimization	Predictive scheduling • Not yard-level, no stacking intelligence	
Portchain	Portchain Yard & terminal planning tools	Yard & terminal planning tools	Visualization and planning depth • Lacks AI-driven rehandle minimization	
AI / Niche Optimization	Inform, Optimization engines for logistics	Optimization engines for logistics	Black-box, limited explainability, • Heavy integration effort	
Kaleris	Kaleris Yard & visibility platforms	Yard & visibility platforms	Focus on execution visibility, not decision optimization	
NextPort	NextPort PCS & ecosystem coordination	PCS & ecosystem coordination	Ecosystem-wide visibility • No yard intelligence or stacking logic	
Your Product	Ctrl+Stack – Yard Optimize			
Strengths	AI-driven yard optimization layer • First-move right placement		New entrant, dependent on data access	
Gaps	• Explainable AI, modular SaaS, TOS-agnostic • New entrant, dependent on data access			



Configurability



Dimension	✗ Before AI	✓ With AI
Decision Approach	Rule-based, FIFO, manual heuristics	AI-driven, predictive optimization
Planning Horizon	Immediate placement only	Looks ahead to future retrievals
Container Placement	First-available or zone-based	Optimal slot minimizing total future moves
Rule Handling	Static, hard-coded rules	Dynamic rules + soft constraints
Reaction to Change	Reactive after congestion occurs	Continuous re-optimization in real time
Reshuffles / Rehandles	High and unavoidable	Significantly reduced
Yard Congestion	Discovered late	Predicted and proactively avoided

Separate Export & Import Zones

Enforce strict separation between export and import containers



Reefer Zone Only

Keep reefer containers in designated power-enabled blocks



DG Segregation Enforcement

Automatically enforce hazardous cargo segregation rules



Minimum DG Distance (meters)

10

Optimization Objectives

Set default weights for optimization goals

Minimize Re-handling



80%

Minimize Dwell Time



60%

Maximize Throughput



70%

Container Stacking Rules



Rule ID	Rule Name	Category	Priority	Rule Strength	Applies Level	Description	Conflict Resolution
CG-01	Same consignee grouping	Commercial Grouping	P2 Productivity	SOFT	STACK	Containers belonging to the same consignee should	Safety > Regulatory > Productivity > Efficiency
CG-02	Same shipping line grouping	Commercial Grouping	P2 Productivity	SOFT	STACK	Containers of same shipping line grouped together	Safety > Productivity Safety > Productivity
CG-03	Same CFS grouping	Operational Grouping	P2 Productivity	SOFT	STACK	Containers from same CFS grouped	Safety > Productivity
DL-01	CFS vs DPD segregation	Flow Segregation	P1 Regulatory	HARD	STACK	DPD and CFS cargo must not be mixed	Safety > Regulatory
FL-01	Export & import segregation		P1 Regulatory	HARD	STACK	Export and import must not be mixed	Safety > Productivity
PD-01	Port of delivery grouping		P1 Regulatory	HARD	STACK	Same POD containers grouped	Safety > Regulatory
SZ-01	Like-on-like stacking	Size Compatibility	P0 Safety	HARD	STACK	Same ISO length stacking only	Safety > Productivity
SZ-02	Like-on-like size stacking	Size Compatibility Weight	P0 Safety	HARD	STACK	Same ISO length stacking	Safety > all
SZ-03	Heavy at bottom		P0 Safety	HARD	STACK	Prevent 40ft on 40ft	Safety > all
WT-01	Heavy at bottom	Weight	P0 Safety	HARD	STACK	Weight decreases by tier	Safety > all
HT-02	DG stacking height limit	Hazmat	P0 Safety	HARD	STACK	Live animals isolated vertically	Safety > all
HZ-19	Weight gradient	Hazmat	P0 Safety	HARD	BLOCK	Reefer containers only tiers	Safety > Productivity
HZ-01	DG quantity calculation	Hazmat	P0 Safety	HARD	BLOCK	Reefer containers only in reefer enabled	Safety > Productivity
MP-01	Minimize rehandle potential	Planning	P2 Productivity	SOFT	STACK	Reefer exports near reefer enabled block	Safety > Productivity
PE-05	RTG travel optimization	Planning	P3 Efficiency	SOFT	BLOCK	Minimize RTG travel distance	Safety > Productivity Productivity > Efficiency

Hazardous Handling Rules

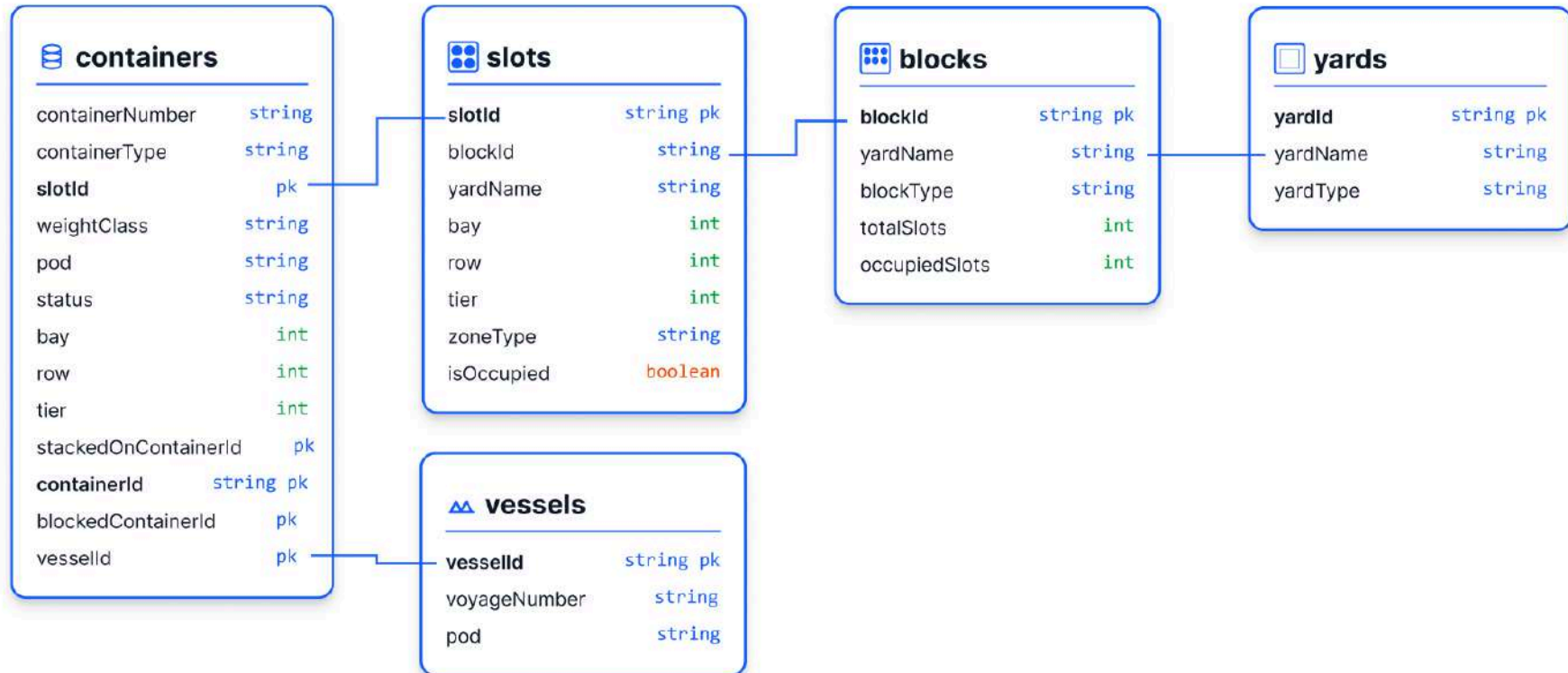


Rule ID	Rule Name	⚙️ Category	⚠️ Priority	📍 Applies	📍 Level
HZ-01	Explosives isolation	Hazmat	P0 Safety	YARD	YARD
HZ-02	Oxidizer vs Flammable Liquid	Hazmat	P1 Regulatory	BAY	BAY
HZ-03	Oxidizer vs flammable solids	Hazmat	P1 Regulatory	BLOCK	BLOCK
HZ-04	Dangerous when wet vs liquids	Hazmat	P0 Safety	ROW	ROW
HZ-05	Toxic gas vs corrosive	Hazmat	P0 Safety	BLOCK	BLOCK
HZ-06	Toxic vs food	Hazmat	P0 Safety	YARD	YARD
HZ-07	Flammable vs corrosive	Hazmat	P1 Regulatory	BAY	BAY
HZ-08	Flammable solid vs oxidizer	Hazmat	P1 Regulatory	ROW	ROW
HZ-09	Flammable gas vs liquid	Hazmat	P1 Regulatory	ROW	ROW
HZ-10	Toxic vs flammable	Hazmat	P1 Regulatory	BAY	BAY
HZ-11	DG vertical incompatibility	Hazmat	P0 Safety	TIER	TIER
HZ-12	DG above food	Hazmat	P0 Safety	TIER	TIER
HZ-13	DG near reefer	Hazmat	P1 Regulatory	BLOCK	
HZ-14	Radioactive isolation	Hazmat	P0 Safety	YARD	
HZ-15	High-risk explosives	Hazmat	P1 Regulatory	BLOCK	

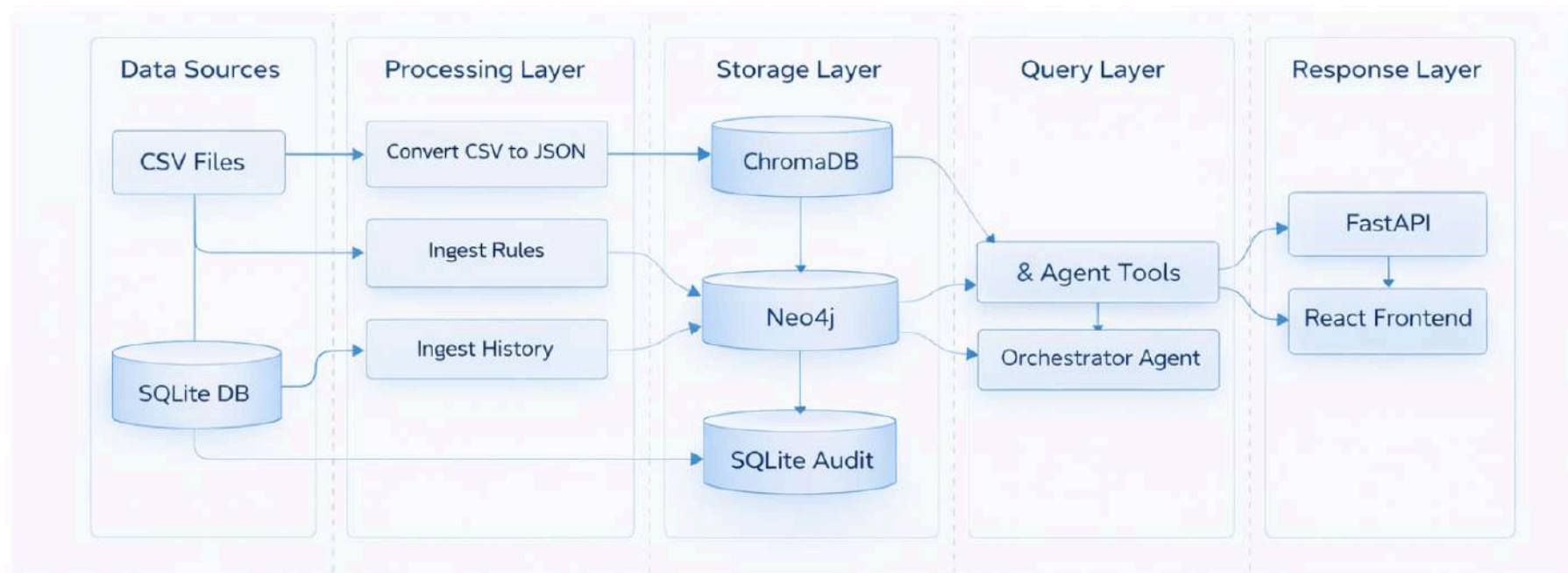
P0 Safety

P1 Regulatory

ER Diagram



Sequence Diagram



Container Management Query Flowchart

